Project : Transient flow modelling
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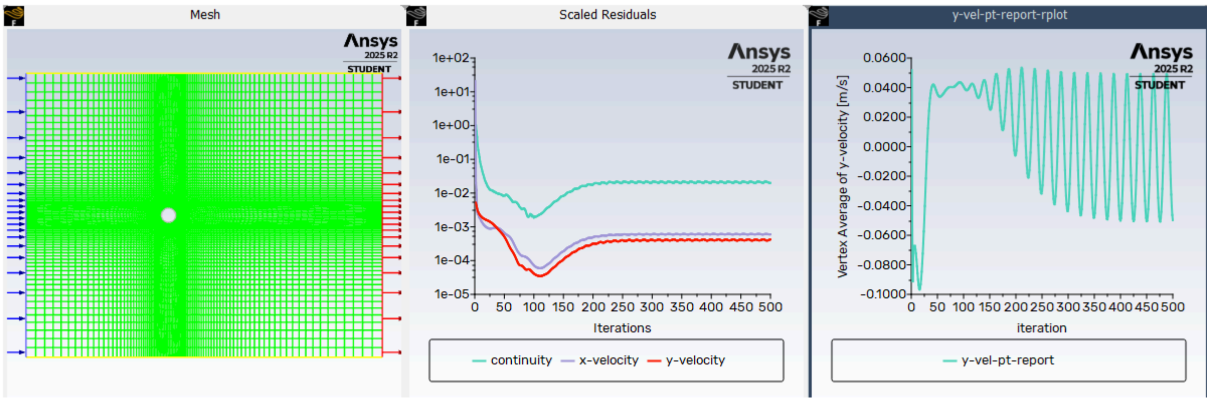
IISC - CFD
–

Steady state flow for reference and baseline calculations. The settings are modified as below after the mesh import as per note 1 and note 2.

Modified Settings Summary

Setting	Current Value	Default Value
Setup		
Models		
Viscous	Laminar	SST k-omega
Cell Zone Conditions		
Boundary Conditions		
Inlet		
in (velocity-inlet, id=4)		
Velocity Magnitude	1 [m/s]	0 [m/s]
Reference values		
Solution		
Methods		
Pressure Velocity Coupling	SIMPLE	Coupled
Controls		

The solution did not converge by using the settings above.

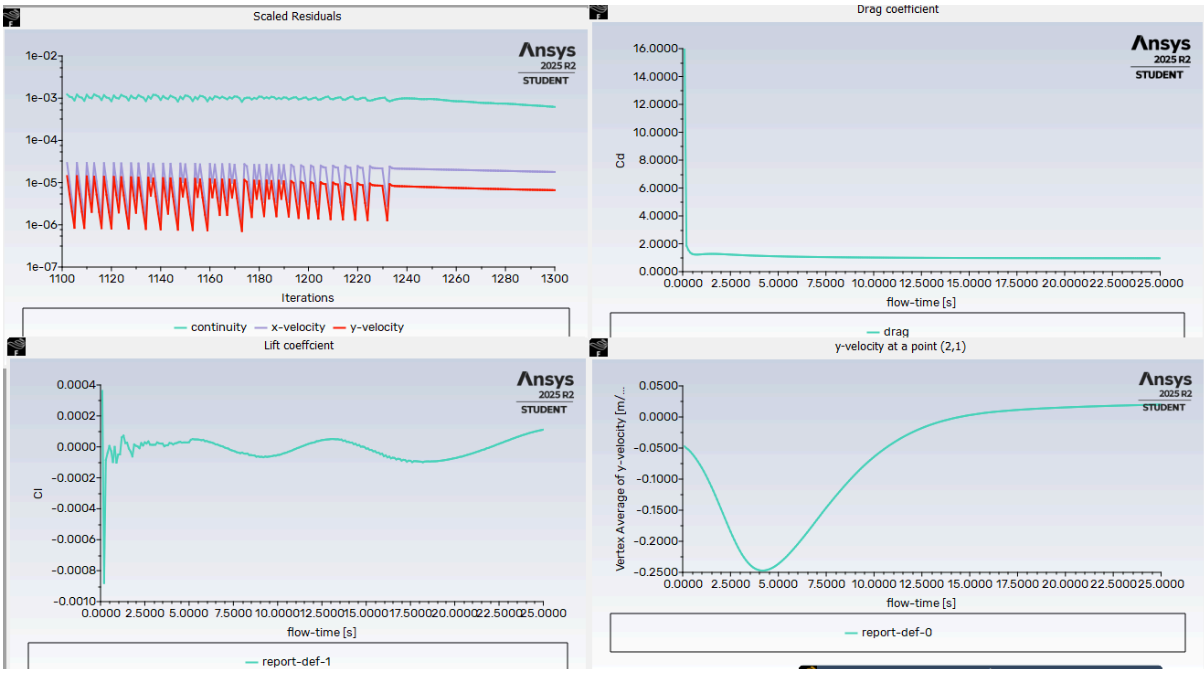


Changing settings for the transient flow as below for the rest of the experiment.

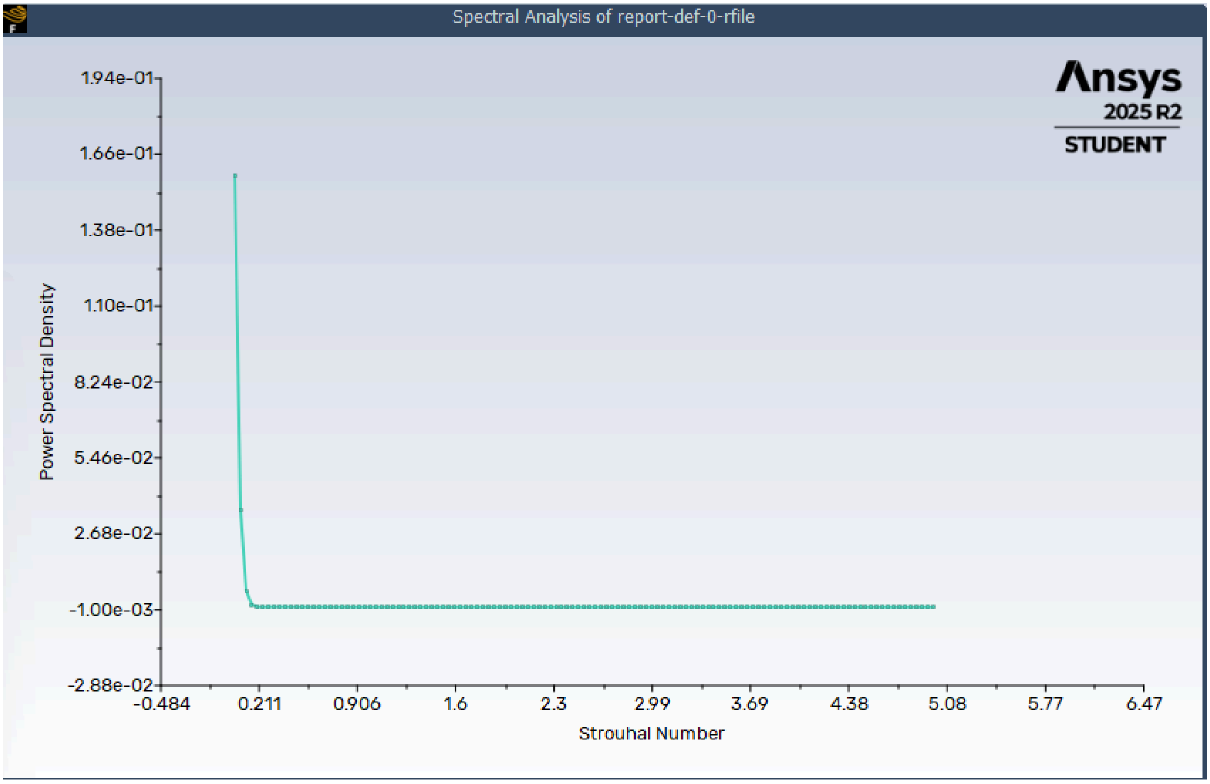
Modified Settings Summary

Setting	Current Value	Default Value
Setup		
Models		
Viscous	Laminar	SST k-omega
Cell Zone Conditions		
Boundary Conditions		
Inlet		
in (velocity-inlet, id=4)		
Velocity Magnitude	1 [m/s]	0 [m/s]
Reference values		
Solution		
Methods		
Transient Formulation	Bounded Second Order Implicit	First Order Implicit
Pressure Velocity Coupling	PISO	SIMPLE
Controls		

Present a 2D plot showing the y-velocity at a point in the wake over at least five periods • Show the drag and lift coefficients in a 2D plot over at least five periods

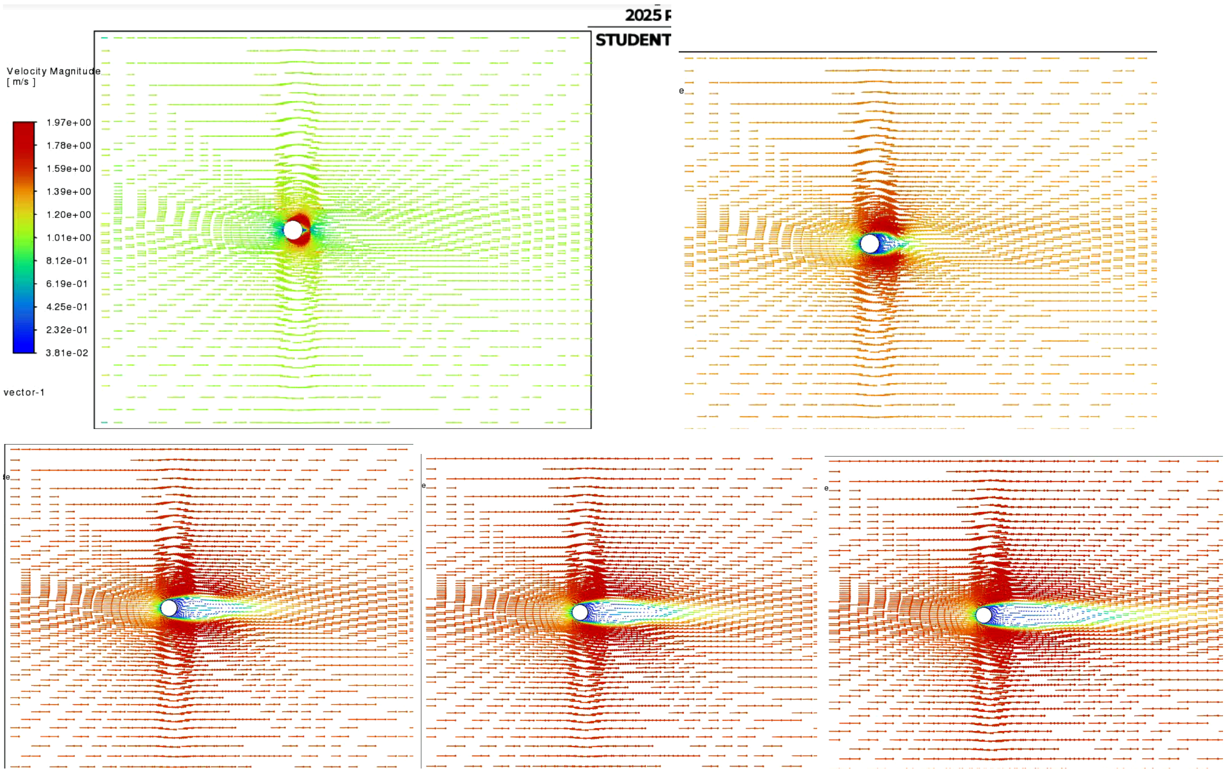


Calculate the Strouhal number using a Fast Fourier Transform of point-data as evidence for the frequency



A single dominant shedding frequency (peak at St ≈ 0.2), indicating periodic vortex shedding in the wake.

Velocity magnitude contour



The velocity contours show alternating vortex shedding in the cylinder wake, indicating unsteady flow.

Transient unsteady flow input summary

```
Models
------

   Model                            Settings
   -------------------------------------------------------------------
   Space                            2D
   Time                             Unsteady, Bounded 2nd-Order Implicit
   Viscous                          Laminar
   Heat Transfer                    Disabled
   Solidification and Melting       Disabled
   Species                          Disabled
   Coupled Dispersed Phase          Disabled
   NOx Pollutants                   Disabled
   SOx Pollutants                   Disabled
   Soot                             Disabled
   Mercury Pollutants               Disabled
   Structure                        Disabled
   Acoustics                        Disabled
   Eulerian Wall Film               Disabled
   Potential/Electrochemistry       Disabled
   Multiphase                       Disabled

Material Properties
-------------------

   Material: air (fluid)

      Property    Units       Method      Value(s)
      ------------------------------------------
      Density     kg/m^3      constant    1
      Viscosity   kg/(m s)    constant    0.01

   Material: aluminum (solid)

      Property    Units     Method     Value(s)
      ---------------------------------------
      Density     kg/m^3    constant   2719
```

```
Cell Zone Conditions
--------------------

   Zones

      name    id   type    material
      -----------------------------
      fluid   2    fluid   air

   Setup Conditions

      fluid

         Condition       Value
         ---------------------
         Material Name   air
         Frame Motion?   #f
         Mesh Motion?    #f
```

```
Boundary Conditions
-------------------

   Zones

      name       id   type
      -------------------------------
      out        3    pressure-outlet
      in         4    velocity-inlet
      sym2       5    symmetry
      sym1       6    symmetry
      cylinder   7    wall

   Setup Conditions

      out

         Condition   Value
         -----------------

      in

         Condition                  Value
         --------------------------------
         Velocity Magnitude [m/s]   1

      sym2

         Condition   Value
         -----------------

      sym1

         Condition   Value
         -----------------

      cylinder

         Condition                  Value
         ------------------------------------------
         Wall Motion                Stationary Wall
         Shear Boundary Condition   No Slip
```

```
Unsteady Calculation Parameters


   ------------------------------
   Number of Time Steps       250
   Time Step Size [s]         0.1
   Max Iterations/Time Step   20

Under-Relaxation Factors

   Variable      Relaxation Factor
   -------------------------------
   Pressure      0.30000001
   Density       1
   Body Forces   1
   Momentum      0.7

Linear Solver

   Variable     Solver Type   Termination Criterion   Residual Reduction Tolerance
   -------------------------------------------------------------------------------
   Pressure     V-Cycle       0.1
   X-Momentum   Flexible      0.1                     0.7
   Y-Momentum   Flexible      0.1                     0.7

Pressure-Velocity Coupling

   Parameter                     Value
   -----------------------------------
   Type                          PISO
   Skewness-Neighbour Coupling   #t
   Skewness Correction           1
   Neighbour Correction          1

Discretization Scheme

   Variable   Scheme
   ------------------------------
   Pressure   Second Order
   Momentum   Second Order Upwind
```